



EE 203: Network Theory

Course Name: EE 203

Course Code: Network Theory

L-T-P-C: 3-0-0-3

Prerequisites: IC 160 Electrical System Around us

Students intended for: UG

Elective or Compulsory: Core for EE, Elective for other UG branches

Approval: 5th Senate, 8th Senate

Course Contents

- **Transient Network Analysis:** Transient and steady state sinusoidal response. Response of RL, RC and RLC networks using Laplace Transforms for unit step, impulse and ramp inputs. [6 Lectures]
- **Two Port Networks and their Characterization:** Open circuit, short circuit, hybrid and transmission parameters; Series, parallel and tandem connections of two-port networks, multi-port networks, multi-terminal networks; Resonant and band pass circuits, magnetically coupled circuits, analysis of coupled circuits. Network transmission criteria; delay and rise time, Elmore's and other definitions. [10 lectures]
- **Network Functions:** Concept of complex frequency, Driving point impedances; Transfer functions of networks, Poles and zeros, Stability analysis. [4 Lectures]
- **Network Synthesis:** Positive real functions and their properties, tests for positive real functions, Hurwitz polynomials; Driving-point synthesis of LC, RC and RL networks, Foster forms and Cauer forms. [8 lectures]
- **Network graphs and their applications in network analysis:** [3 lectures]
- **Three-Phase A.C. Circuit Analysis:** Analysis of balanced and unbalanced three-phase networks; Symmetrical components and their application in analysis of unbalanced networks. [3 Lectures]
- **Analysis of A.C. circuits with non-sinusoidal inputs:** Filters: Introduction to filters, various types of filters - LP, HP, BP and BS. Transformation of LP to other types. Butterworth, Chebyshev and Elliptic approximations to LPF. [2 Lectures]
- **Frequency response:** Polar plots, magnitude and phase plots, Bode plot. [6 lectures]

Text & Reference Books:

1. Franklin Fa-Kun Kuo, **Network Analysis and Synthesis**, John Wiley & Sons, 1996.
2. Van Valkenburg, **Network Analysis**, PHI Learning, 2014.